
eGamma-VBA: expanded Gamma distribution for Microsoft Excel

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ABOUT

1.1 Introduction

The expanded gamma distribution extends the traditional three-parameter gamma distribution to also accommodate left-skewed data. Its versatility is particularly beneficial in project cost estimation and other fields where such data patterns are common. Although the distribution has been in de facto use in Scandinavian cost-estimation practice since the 1970s, it has only recently been formally defined. The *EGAMMA-VBA* library provides a Python implementation of the expanded gamma distribution, along with the tools for its practical application.

The *EGAMMA-VBA* library enhances builds on implementation of the gamma distribution, enhancing it to incorporate the specialized attributes of the expanded version. Additionally, *EGAMMA-VBA* introduces capabilities beyond those offered by SciPy, such as three-point estimation. This method, used with expert judgment, determines the most likely, optimistic, and pessimistic scenarios. Unlike standard practice, which often places optimistic and pessimistic estimates at the 10th and 90th percentiles, *EGAMMA-VBA* offers the flexibility to choose any percentile range, allowing for customized three-point estimation according to user preference.

1.2 Documentation Content

- **About:** The current section.
- **Mathematical Definitions:** Mathematical underpinnings of the distribution's functions and statistical measures.
- **Installation:** Instructions for installing the library.
- **Usage:** Introduction to the library's basic usage.
- **Functions:** In-depth descriptions of the library's functions, including parameters and return values.

1.3 Support

If you encounter any problems or have any questions, please open an issue on the [GitHub repository issue tracker](#).

1.4 Contact Information

For inquiries or collaboration on the *EGAMMA-VBA* library, contact:

- **Associate Professor Frode Drevland**
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Dr. Drevland is dedicated to the continuous development of the *EGAMMA-VBA* library and welcomes feedback, suggestions, and contributions from the community.

1.5 License

This library is distributed under the MIT License. See [LICENSE](#) for more information.

MATHEMATICAL DEFINITIONS

2.1 Introduction

This page is the reference for the mathematics behind the library: the density and its cumulative distribution function, the statistical measures, and the procedure by which the parameters are recovered from a three-point estimate. Nothing here is needed to use the library; see [Usage](#) for that.

2.2 Parameters

The distribution is characterised by three parameters:

- α (alpha): The shape parameter, which must be positive ($\alpha > 0$).
- β (beta): The scale parameter, which must be non-zero ($\beta \neq 0$). Its sign determines the direction of skewness: positive values give right skew, negative values left skew.
- δ (delta): The location parameter, which may be any real number. It is the bound of the support, lower when $\beta > 0$ and upper when $\beta < 0$.

The support is $[\delta, \infty)$ for a positive scale and $(-\infty, \delta]$ for a negative one.

The distribution is strictly skewed in one direction or the other; exact symmetry is not attainable at any finite α . As α grows the shape converges to that of a normal distribution, so a large value such as 10^9 approximates a symmetric case closely.

2.3 Distribution Functions

The probability density function (PDF) and cumulative distribution function (CDF) are defined as follows.

Probability Density Function (PDF):

$$f(x; \alpha, \beta, \delta) = \begin{cases} \frac{|x - \delta|^{\alpha-1} e^{-(x-\delta)/\beta}}{|\beta|^\alpha \Gamma(\alpha)}, & \text{for } \frac{x - \delta}{\beta} > 0 \\ 0, & \text{otherwise} \end{cases}$$

where $\Gamma(\alpha)$ is the gamma function $\int_0^\infty t^{\alpha-1} e^{-t} dt$

Cumulative Density Function (CDF):

$$F(x; \alpha, \beta, \delta) = \begin{cases} \frac{\gamma(\alpha, \frac{x-\delta}{\beta})}{\Gamma(\alpha)}, & \text{for } \beta > 0, \frac{x-\delta}{\beta} > 0 \\ 1 - \frac{\gamma(\alpha, \frac{x-\delta}{\beta})}{\Gamma(\alpha)}, & \text{for } \beta < 0, \frac{x-\delta}{\beta} > 0 \end{cases}$$

where $\gamma(\alpha, \frac{x-\delta}{\beta})$ is the lower incomplete gamma function $\int_0^{\frac{x-\delta}{\beta}} t^{\alpha-1} e^{-t} dt$.

The CDF has no closed form, so the percent point function must be obtained numerically. That single fact shapes the fitting procedure below.

2.4 Statistical measures

Expected value:

$$E[X] = \alpha\beta + \delta$$

Variance:

$$\text{Var}(X) = \alpha\beta^2$$

Skewness:

$$\text{Skew}(X) = \frac{\beta}{|\beta|} \times \frac{2}{\sqrt{\alpha}}$$

Excess kurtosis:

$$\text{ExKurt}(X) = \frac{6}{\alpha}$$

Mode, for $\alpha > 1$:

$$\text{Mode}(X) = (\alpha - 1)\beta + \delta$$

For $\alpha \leq 1$ the density is monotone on its support and the mode lies at δ .

2.5 Fitting to a three-point estimate

Given elicited values $L \leq M \leq H$, where L and H are the P_L and $P_H = 1 - P_L$ percentiles and M is the mode, the problem is to find α , β and δ reproducing all three.

Writing $G^{-1}(p; \alpha)$ for the percent point function of the standard gamma distribution with unit scale and zero location, and taking $\beta > 0$ for the moment, the ratio of the two half-ranges eliminates both the scale and the location and leaves a single equation in the shape parameter:

$$\frac{M - L}{H - M} = \frac{(\alpha - 1) - G^{-1}(P_L; \alpha)}{G^{-1}(P_H; \alpha) - (\alpha - 1)}$$

This has no closed-form solution and is solved numerically. The library bisects on skewness, $2/\sqrt{\alpha}$, rather than on α itself, since skewness is bounded on $(0, 2]$ and needs no interval derived from the ceiling.

Excel's `GAMMA.INV` is the numerical foundation for all of this, and its accuracy at very large shape parameters is the practical reason the ceiling is set where it is.

With α known, the remaining parameters follow in closed form. The scale comes from the full elicited span, which is better conditioned than either half:

$$\beta = \frac{H - L}{G^{-1}(P_H; \alpha) - G^{-1}(P_L; \alpha)}$$

and the location from the mode relation:

$$\delta = M - (\alpha - 1)\beta$$

A left-skewed estimate, one for which $M - L > H - M$, is the mirror image. It is solved by inverting the ratio, solving the same equation, and negating the scale before computing the location.

2.6 Admissible estimates

Not every ordering $L \leq M \leq H$ can be fitted. The mode of the fitted distribution cannot fall outside the interval the elicited percentiles enclose, and requiring $L \leq M$ therefore requires $G^{-1}(P_L; \alpha) \leq \alpha - 1$. The shape at which the two coincide is the smallest admissible one. For $P_L = 0.10$ it is $\alpha = 1.15631$, corresponding to a skewness of 1.85991, so no valid estimate implies more skewness than that. The bound depends on P_L ; a wider elicited interval admits more.

The boundary at $\alpha = 1$, where the mode reaches the support endpoint, lies strictly inside the inadmissible region and is never reached from a valid estimate.

A symmetric estimate is reproducible only in the limit $\alpha \rightarrow \infty$. The library returns a ceiling of 10^9 , which reproduces the elicited values to about 1.5×10^{-5} of the elicited range rather than exactly.

2.7 Accuracy

For any admissible asymmetric estimate, the deviation of a recovered elicited value, normalised by the elicited range, is bounded by

$$E < \frac{\tau}{2(2 - \tau)} = \frac{\tau}{4} + O(\tau^2)$$

where τ is the relative acceptance threshold on the ratio. At the default $\tau = 10^{-10}$ this is 2.5×10^{-11} . The bound holds for every admissible input rather than for tested cases, and it excludes the symmetric branch, where the ceiling rather than the threshold governs.

Recovery of α itself is a weaker guarantee. The ratio above approaches unity as $\alpha^{-1/2}$, so the recovery becomes increasingly ill-conditioned near symmetry and a fixed threshold buys progressively less accuracy in the shape parameter. This is a property of the problem and not of the solver; tightening the threshold improves the recovered shape at the cost of more iterations.

INSTALLATION

To integrate the EGAMMA library into Excel, you have two distinct installation methods to choose from, depending on your needs and preferences.

Installing as an .xlam Add-in for Global Access If you require the EGAMMA library to be accessible across all Excel workbooks on your system, the optimal approach is to install it as an .xlam add-in. This method ensures that the library's features are readily available in Excel, no matter which workbook you open.

Importing VBA Code for Workbook-Specific Use Alternatively, if your intention is to utilize the EGAMMA library in only a few selected workbooks, you may opt to import the library's code directly into those workbooks via Excel's VBA code editor. This approach allows for a more targeted use of the library, without affecting the global Excel environment.

For comprehensive guidance on executing these installation methods, please refer to the following instructional pages, which offer step-by-step procedures tailored to each approach.

3.1 Install the EGAMMA.xlam Plugin for Excel

This page provides comprehensive instructions for installing the EGAMMA.xlam plugin, enabling the use of EGAMMA functions universally across all Excel workbooks. For a more selective approach, you have the option to import the EGAMMA code into individual workbooks as needed (refer to *Importing the EGAMMA VBA Code into an Excel Workbook* for details).

To equip your Excel environment with the EGAMMA.xlam plugin, please proceed with the following download and installation steps:

3.1.1 Step 1: Download the Plugin

- Visit the [EGAMMA-VBA GitHub Releases page](<https://github.com/FrodeDrevland/EGAMMA-VBA/releases/latest>)
- Click on the download link for the EGAMMA.xlam file.
- Save the file to a known location on your computer, such as your Desktop or a dedicated Downloads folder.

3.1.2 Step 2: Open Excel

- Launch Microsoft Excel on your computer.

3.1.3 Step 3: Access the Add-ins Menu

- For Excel 2010 and later: - Go to the File tab, click on Options, and then select Add-Ins. - At the bottom of the window, in the Manage box, select Excel Add-ins and click Go.
- For Excel 2007: - Click the Microsoft Office Button, then click Excel Options, and choose Add-Ins. - At the bottom of the window, in the Manage box, select Excel Add-ins and click Go.

3.1.4 Step 4: Install the Plugin

- In the Add-Ins window, click Browse . . .
- Navigate to the location where you saved the EGAMMA.xlam file.
- Select the file and click OK.
- Ensure that the checkbox next to EGAMMA is checked in the list of available Add-Ins.
- Click OK to close the Add-Ins window.

3.1.5 Step 5: Confirm Installation

- The EGAMMA plugin should now be installed and ready to use.
- Confirm the installation by looking for by checking that the EGAMMA functions work (See [Usage](#) documentation).

3.1.6 Step 6: Enable Macros if Necessary

- The plugin relies on includes macros. If the formulae are not working, you may need to enable macros: - Go to the File tab, click on Options, then select Trust Center, and click on the Trust Center Settings . . . button. - In the Trust Center, select Macro Settings and choose the appropriate setting to enable macros. Be cautious and only enable macros from trusted sources.

3.1.7 Troubleshooting

- If the EGAMMA functions does not appear, restart Excel and try again.
- If you encounter any issues, check that your version of Excel supports .xlam plugins and that you have the necessary permissions to install add-ins on your system.

3.2 Importing the EGAMMA VBA Code into an Excel Workbook

This guide outlines the steps to integrate the EGAMMA VBA code into an individual Excel workbook, which allows you to utilize EGAMMA's capabilities within that specific file. If you wish to extend EGAMMA's functionalities to all workbooks on your system, consider installing the EGAMMA.xlam plugin instead (see [Install the EGAMMA.xlam Plugin for Excel](#) for installation guidance).

To begin using the EGAMMA VBA code within a single workbook, you will need to download the EGAMMA.bas file. This file is a VBA module containing the necessary code to be imported into the VBA editor of Excel. Doing so enables you to tailor and engage with the EGAMMA features directly in your chosen Excel workbook.

Please follow the detailed steps below to import the EGAMMA.bas file into your workbook:

3.2.1 Step 1: Download the EGAMMA.bas File

- Visit the [EGAMMA-VBA GitHub Releases page](<https://github.com/FrodeDrevland/EGAMMA-VBA/releases/latest>) to access the latest version of the EGAMMA.bas file.
- Click on the link for the EGAMMA.bas file to initiate the download.
- Save the file to a familiar directory on your computer for easy retrieval.

3.2.2 Step 2: Enable the Developer Tab in Excel

- For Excel 2010 and newer: - Navigate to the File tab, select Options, and then choose Customize Ribbon. - In the customization area, ensure the Developer checkbox is selected and confirm with OK.
- For Excel 2007: - Click the Microsoft Office Button, select Excel Options, and within the Popular category, activate the Show Developer tab in the Ribbon option.

3.2.3 Step 3: Access the VBA Editor

- From the Developer tab, initiate the Visual Basic command to open the VBA editor.

3.2.4 Step 4: Import the .bas File

- Within the VBA editor, right-click on the VBAProject pertaining to your workbook.
- Choose Import File... and navigate to the directory where the EGAMMA.bas file is located.
- Select the file and opt for Open to import it into your project.

3.2.5 Step 5: Save Your Workbook with Macros

- Return to Excel and save your workbook by selecting Save.
- If your workbook is in .xlsx format, you'll need to switch to the .xlsm format to preserve the VBA code.

3.2.6 Step 6: Enable Macros if Necessary

- To execute the VBA code, ensure that macros are enabled by visiting the File tab, selecting Options, followed by Trust Center, and then Trust Center Settings....
- In the Macro Settings section, choose a suitable option to enable macros, keeping security in mind.

3.2.7 Troubleshooting Tips

- If the Developer tab or VBA editor is not visible, they may need to be enabled in your Excel settings.
- Should you face any issues with the code execution, verify the compatibility of the code with your Excel version and check for required references.

Always back up your data before importing a .bas file, particularly if it includes extensive macros or code that could modify your data significantly.

USAGE

Once the EGAMMA plugin is installed, its functions become accessible within your Excel workbook(s). This section offers a general overview of how to use these functions.

4.1 EGAMMA_DIST and EGAMMA_INV Functions

The EGAMMA_DIST and EGAMMA_INV functions are analogous to Excel's native GAMMA.DIST and GAMMA.INV functions, respectively. The key enhancements in the EGAMMA versions include an additional *delta* parameter for location adjustment, and the ability for the *beta* parameter to accept negative values, accommodating negatively skewed distributions.

For instance, to calculate the cumulative probability for a value of 200 in an expanded distribution characterized by $\alpha = 10$, $\beta = 20$, and $\delta = 100$, use:

```
=EGAMMA_DIST(200, 10, 20, 100, TRUE)
```

4.2 Statistical Measures

The EGAMMA library includes a range of functions designed to compute various statistical measures based on the parameters of the expanded gamma distribution (α , β , and δ). These functions operate similarly to standard Excel functions:

- EGAMMA_MEAN(α , β , δ): Calculates the mean.
- EGAMMA_MODE(α , β , δ): Calculates the mode.
- EGAMMA_MEDIAN(α , β , δ): Calculates the median.
- EGAMMA_VAR(α , β): Calculates the variance.
- EGAMMA_STDDEV(α , β): Calculates the standard deviation.
- EGAMMA_SKEW(α , β): Calculates the skewness.
- EGAMMA_KURT(α): Calculates the excess kurtosis.

These are functions of the distribution's parameters alone, and apply equally whether the parameters came from a three-point estimate, from a fit to data, or were entered directly.

4.3 Parameter estimation

The EGAMMA library offers two distinct functions for parameter estimation: EGAMMA_TPE_TO_PARAMS and EGAMMA_FIT_TO_PARAMS. These functions are designed to calculate the parameters of the Expanded Gamma Distribution either from a three-point estimate or by fitting to a data set.

4.3.1 EGAMMA_TPE_TO_PARAMS

The EGAMMA_TPE_TO_PARAMS function requires three key values as arguments: the low, most likely, and high values of a three-point estimate. Additionally, it accepts an optional low probability value, which is the probability of the low estimate (defaulting to 0.1). It's important to note that the probability of the high estimate is always the complement of the low probability ($1 - \text{low probability}$).

```
=EGAMMA_TPE_TO_PARAMS(low, most-likely, high, [low-probability])
```

```
Example: =EGAMMA_TPE_TO_PARAMS(100, 200, 400, 0.1)
```

The function outputs a 3x1 array: alpha in the function's cell, and beta and delta in the two adjacent cells to the right. If these adjacent cells are not empty, the function will return #SPILL!, indicating it cannot display the results.

The fitted parameters recover the three elicited values to within 2.5×10^{-11} of the elicited range. The exception is an estimate too close to symmetry to resolve: a perfectly symmetric estimate corresponds to an infinite shape parameter, so those are fitted at the library's finite ceiling and recover the elicited values to about 1.5×10^{-5} of the range instead. EGAMMA_TPE_AT_CEILING(low, most-likely, high, [low-probability]) returns TRUE for such an estimate and FALSE for one fitted to the tolerance.

A most likely value that equals, or nearly equals, one of the outer values is admissible and is fitted like any other estimate; the endpoint case is not treated separately. An estimate is rejected with #N/A only if the three values are not ordered, that is unless $\text{low} \leq \text{most-likely} \leq \text{high}$ and $\text{low} < \text{high}$.

4.3.2 EGAMMA_FIT_TO_PARAMS

EGAMMA_FIT_TO_PARAMS is designed to estimate the alpha, beta, and delta parameters of the Expanded Gamma Distribution from a given data set. This function uses the method of moments, employing the skewness of the data to estimate alpha, the standard deviation for beta, and the mean, alpha, and beta to calculate delta.

```
=EGAMMA_FIT_TO_PARAMS(range)
```

```
Example: =EGAMMA_FIT_TO_PARAMS(A1:A100)
```

Similar to EGAMMA_TPE_TO_PARAMS, the results are returned as an array. The alpha parameter appears in the cell where the function is entered, with beta and delta in the adjacent right cells.

For further details, consult the [Functions](#) documentation.

FUNCTIONS

EGAMMA_DIST(*x*, alpha, beta, delta, cumulative):

Calculate the expanded gamma distribution.

Parameters

- **x** – Required. The value at which the function is evaluated.
- **alpha** – Required. The shape parameter of the distribution.
- **beta** – Required. The scale parameter of the distribution. Negative beta entails a left skewed distribution.
- **delta** – Required. The location parameter of the distribution.
- **cumulative** – Required. A logical value that determines the form of the function. If cumulative is TRUE, GAMMA.DIST returns the cumulative distribution function; if FALSE, it returns the probability density function.

Returns

The value of the (cumulative) gamma distribution.

Return type

float

EGAMMA_INV(probability, alpha, beta, delta):

Calculate the inverse of the expanded gamma distribution.

Parameters

- **probability** – The probability for which the inverse is calculated.
- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns

The inverse value of the expanded gamma distribution for the given probability.

EGAMMA_MEAN(alpha, beta, delta):

Calculate the mean of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

- **delta** – The location parameter of the expanded gamma distribution.

Returns

The mean of the expanded gamma distribution.

EGAMMA_MODE(alpha, beta, delta):

Calculate the mode of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns

The mode of the expanded gamma distribution.

EGAMMA_MEDIAN(alpha, beta, delta):

Calculate the median of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns

The median of the expanded gamma distribution.

EGAMMA_VAR(alpha, beta):

Calculate the variance of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

Returns

The variance of the expanded gamma distribution.

EGAMMA_STDDEV(alpha, beta):

Calculate the standard deviation of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

Returns

The standard deviation of the expanded gamma distribution.

EGAMMA_SKEW(alpha, beta):

Calculate the skewness of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.

- **beta** – The scale parameter of the expanded gamma distribution.

Returns

The skewness of the expanded gamma distribution.

EGAMMA_KURT(alpha):

Calculate the excess kurtosis of the expanded gamma distribution.

Parameters

alpha – The shape parameter of the expanded gamma distribution.

Returns

The excess kurtosis of the expanded gamma distribution.

EGAMMA_TPE_AT_CEILING(low, likely, high, low_probability=0.1):

Report whether a three-point fit met its tolerance or returned the shape ceiling instead.

A perfectly symmetric estimate is only reproducible in the limit as the shape tends to infinity, so estimates too near symmetry to resolve are fitted at the library's finite ceiling. Those fits reproduce the elicited values to about 1.5×10^{-5} of the elicited range rather than to the tolerance of 2.5×10^{-11} . Returning the ceiling silently would leave no way to tell the two cases apart.

Parameters

- **low** – The low estimate of the distribution.
- **likely** – The most likely estimate of the distribution.
- **high** – The high estimate of the distribution.
- **low_probability** – The probability of the low estimate (default is 0.1).

Returns

TRUE if the fit returned the shape ceiling, FALSE if it met the tolerance.

EGAMMA_FIT_TO_PARAMS(*args):

Fit the parameters of the expanded gamma distribution to a given set of data points.

This function estimates the alpha, beta, and delta parameters of the expanded gamma distribution, based on the provided data using the method of moments. It uses the skewness of the data to estimate alpha, then calculates beta using the standard deviation, and finally computes delta from the mean, alpha, and beta.

Parameters

args – A variable number of arrays or lists containing numerical data points. These are combined into a single dataset for the analysis.

Returns

A list containing the estimated parameters [alpha, beta, delta] of the extended gamma distribution.

Raises

ValueError – If the input data does not allow for a proper fit, e.g., if it leads to divide-by-zero errors.

EGAMMA_TPE_TO_PARAMS(low, likely, high, low_probability=0.1):

Estimate the parameters of the expanded gamma distribution based on a three-point estimate.

This function uses a low, likely, and high estimate to determine the parameters of the extended gamma distribution.

The shape parameter is found by bisection on the magnitude of the skewness. The quantity matched is the mode's position within the elicited range,

$$t = \frac{\min(\text{likely} - \text{low}, \text{high} - \text{likely})}{\text{high} - \text{low}},$$

against the position the standard Gamma with that shape produces between its two percentiles,

$$t_{\alpha} = \frac{\alpha - 1 - q_L}{q_H - q_L},$$

and the search stops when the two differ by less than 2.5×10^{-11} . That difference is itself the largest error, as a fraction of the elicited range, with which the three elicited values are recovered from the fitted parameters, so the tolerance is stated directly in the quantity of interest. A most likely value equal to an outer value gives $t = 0$ and goes through the same search; there is no separate case for it.

An estimate too near symmetry to resolve is fitted at the shape ceiling instead; use EGAMMA_TPE_AT_CEILING to detect this.

Parameters

- **low** – The low estimate of the distribution.
- **likely** – The most likely estimate of the distribution.
- **high** – The high estimate of the distribution.
- **low_probability** – The probability of the low estimates (default is 0.1).

Returns

A list containing the estimated parameters [alpha, beta, delta] of the extended gamma distribution.

Raises

ValueError – Returns #N/A if the input values are inconsistent (low > likely, high < likely, or low >= high), #NUM! if the low probability is outside (0, 0.5) or the search cannot meet its tolerance.